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RectMamba: Exploring state space models with entropy-divergence framework for noisy label rectification

Ningwei Wang[a,1], Weiqiang Jin[a,1], Haixia Bi[a,*], Guang Yang[b,c,d,e]

[a] School of Information and Communications Engineering, Xi'an Jiaotong University, Xi'an, Shaanxi, 710049, Shaanxi, China
[b] Bioengineering Department and Imperial-X, Imperial College London, London, W12 7SL, UK
[c] National Heart and Lung Institute, Imperial College London, London, SW7 2AZ, UK
[d] Cardiovascular Research Centre, Royal Brompton Hospital, London, SW3 6NP, UK
[e] School of Biomedical Engineering & Imaging Sciences, King's College London, London, WC2R 2LS, UK

HIGHLIGHTS

- Convolutional Visual Mamba (CVMamba): CVMamba utilizes the hierarchical feature representations produced by the VSS blocks to refine noisy labels progressively. By leveraging the selective scan capabilities of the SS2D module, the corrected labels align more accurately with the true underlying patterns in the data, thus enhancing the model's overall performance and resilience against label noise.
- Entropy-Divergence K-Nearest Neighbors Relabel (EDKR): EDKR corrects labels by combining Entropy-Divergence with KNN. It extracts features using a pre-trained network, calculates prediction probabilities, and uses divergence to assess confidence. Low-entropy, high-confidence samples are relabeled. KNN evaluates sample proximity, normalizes similarity scores, and identifies clean samples based on consistency for improved training.
- AdaptiveMix Data Augmentation: AdaptiveMix addresses class imbalances by dynamically adjusting instance counts to match the most populous class, ensuring balanced model training. Unlike traditional Mixup, AdaptiveMix refines the mixup coefficient from a Beta distribution, adjusting augmentation intensity during training. Our method enhances model generalization by effectively using both clean and noisy data, training the encoder with cross-entropy loss on mixed data.

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ABSTRACT

The efficacy of Deep Neural Networks (DNNs) is significantly influenced by the quality of labels. Noisy labels in real-world datasets can mislead DNNs into learning incorrect patterns. Existing "learning with noisy labels" (LNL) methods often struggle in high-noise environments, where the network tends to overfit noisy labels during the initial training period, leading to suboptimal performance. Additionally, they tend to discard noisy samples too aggressively, resulting in the loss of valuable information. Moreover, many of these methods exhibit a bias toward selecting simpler, easier-to-classify instances, which causes them to overlook more complex but informative samples, further limiting their effectiveness. To address these limitations, we introduce a robust label correction mechanism within CVMamba. This mechanism refines noisy labels using hierarchical feature representations from Visual State Space (VSS) blocks and the 2D-Selective-Scan (SS2D) module, thereby enhancing model performance and resilience against label noise. Additionally, we propose two innovative components: Entropy-Divergence K-Nearest Neighbors Relabel (EDKR) and Adaptivemix data augmentation. The Entropy-Divergence method, a parameter-free label correction strategy, leverages feature similarity and entropy-divergence uncertainty assessment to accurately correct mislabeled samples. Furthermore, the KNN method evaluates the proximity of samples in the feature space, assigns normalized weights to the nearest neighbors, and utilizes a score matrix with a set threshold to select clean samples. Adaptivemix, enhanced by a refined mixup strategy, adjusts the mixing coefficient based on the training stage to optimize the combination of sample pairs and their labels. This integrated approach has demonstrated substantial improvements in label accuracy, increasing from 20 % to an average of 96 % on the CIFAR-10 dataset with 80 % symmetric label noise. Moreover, our method achieves test accuracies of 82.1 % on CIFAR-100 with 40 % asymmetric label noise, surpassing the previous SOTA accuracy

* Corresponding author.
Email addresses: ningweiwang@stu.xjtu.edu.cn (N. Wang), weiqiangjin@stu.xjtu.edu.cn (W. Jin), haixia.bi@xjtu.edu.cn (H. Bi), g.yang@imperial.ac.uk (G. Yang).

[1] Both authors, N.W. and W.J., contributed equally to this work.

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of 78.0 %. On ANIMAL-10N, we achieve an accuracy of 92.5 %, exceeding the previous best of 90.5 %. On Clothing1M, our method reaches 74.93 % accuracy, nearing the SOTA performance of 75.31 %. These results illustrate RectMamba's effectiveness in managing noisy datasets and highlight its potential in improving DNN robustness against label noise. Code is available at https://github.com/ningwei-wang/RectMamba

1. Introduction

DNNs are highly effective in various tasks, such as image recognition [1,2], object detection [3], image classification [4,5], visual tracking [6], and image segmentation [7]. However, their performance heavily depends on the quality of label annotations, since DNNs tend to memorize noisy labels, which can lead to suboptimal learning outcomes [8]. In real-world scenarios, it is common to encounter training samples with noisy labels due to factors such as human error, inaccurate labeling techniques, and indistinguishable edge mixing.

Learning with Noisy Labels (LNL) has been the subject of many recent studies. A typical approach begins with a "warm-up" stage, brief supervised training on a noisily labeled dataset, before applying more advanced algorithms to handle label noise [9–11]. During this warm-up phase, the network's inherent resilience to noise typically reduces the classification loss of accurately labeled samples faster than that of noisy ones. Although often overlooked in prior studies due to its algorithmic simplicity, this stage is critical to the success of LNL. Unfortunately, it is prone to memorizing noise, making its effectiveness dependent on the noise level, the number of training iterations, and other hyperparameters.

With the advancement of deep learning, various models have emerged as robust tools for feature extraction during the warm-up phase. Traditional approaches often utilize Convolutional Neural Networks (CNNs) [1,12–14] or Vision Transformers (ViTs) [15–17]. CNNs are praised for their scalability and straightforward computational complexity, but their small receptive fields lead to a local bias. Moreover, the uniform application of convolutional kernels across input regions limits their adaptability to novel or low-quality images. In contrast, ViTs enhance visual representation by leveraging global receptive fields and adaptive weights, but suffer from quadratic computational complexity as input size increases. Efforts to optimize this, such as reducing dimensionality or adjusting stride in processing windows, often compromise the extent of receptive fields.

To address the limitations of CNNs and ViTs, Selective Structured State Space Models (SSMs), also known as Mamba, have emerged as a promising solution. Mamba models provide comprehensive global receptive field coverage and employ dynamic weights with linear complexity, making them highly effective for handling long sequences in Natural Language Processing (NLP) tasks [18]. Building on the success of Mamba in NLP, Visual Mamba adapts this architecture for computer vision by integrating a selective scan module, which improves processing efficiency and visual data analysis [19]. This transition from Mamba to Visual Mamba marks a strategic shift toward developing more specialized models that are both robust and versatile across diverse vision domains [20–23]. Furthermore, pre-training the Visual Mamba on the ImageNet dataset minimizes reliance on external data and reduces the risk of noise memorization during the initial training phase. This approach leverages a discriminative backbone network to effectively distinguish between correct and incorrect labels.

Considering these potential vulnerabilities associated with the warm-up phase, several strategies have been proposed to enhance the robustness of algorithms against LNL. This leads to a discussion of two primary strategies for addressing the challenge of noisy labels [24,25]. The first strategy focuses on the identification and removal of erroneous labels prior to model training [26–29]. Such identification can be achieved through techniques such as manual inspection, clustering, or automated outlier detection algorithms. However, a potential drawback of these methods is the need for an additional threshold to correct noisy labels. Additionally, they may lead to the loss of valuable data if noisy labels are discarded without considering their informative content.

Our proposed method involves an initial encoding of data to extract features, on which a classifier determines labels. These predictions are refined using softmax to yield class probabilities. Our approach innovatively leverages both entropy and the margin between the highest and second-highest probabilities to assess the confidence level of each prediction. The heatmap in Fig. 1a demonstrates this process, where darker blue areas signify higher confidence and redder areas indicate greater uncertainty. Accurate predictions align with the lighter diagonal, while dark off-diagonal areas signal potential misclassifications. Predictions with notably low Entropy-Divergence, indicating high confidence, are used to correct corresponding noisy labels, a process that has been experimentally shown to be effective. As illustrated in Fig. 1b, our approach increased label accuracy on the CIFAR-10 dataset from an initial 20 % to an average of 96 %. Notably, the accuracy for the *Automobile* category reached 99 %.

The second method directly trains models to be resilient to noise on corrupted datasets [30–32]. Earlier techniques often reduce noise impact by training with a clean subset of the data [9,30,33]. These methods leverage ensemble predictions to improve overall performance, allowing the models to focus on trustworthy examples while ignoring noisy ones. Despite advancements in these techniques, the inherent bias toward selecting easier classes during the sample selection process remains a key challenge. This bias can lead to the neglect of complex yet informative samples, thus undermining the model's overall accuracy and effectiveness. To overcome this limitation and improve noise robustness during training, our approach adopts a more nuanced sample selection strategy. It leverages the feature space to evaluate sample proximity and interrelationships.

While the above methods prioritize reliable examples and primarily focus on training with a clean subset of data, recent techniques present a shift in perspective. These newer approaches integrate semi-supervised learning with advanced strategies like Mixup [34], MetaMixUp [35], and CutMix [36]. While these methods have expanded the use of both labeled and unlabeled data, they often rely on static data augmentation strategies, which may not optimally exploit the dynamic nature of training datasets. Our method addresses this limitation by dynamically fine-tuning the data augmentation intensity as the training progresses. In this paper, we propose the Mamba with Entropy-Divergence Framework for Label Noise Rectification (RectMamba) to address these challenges. Its core concepts are summarized as follows:

- Convolutional Visual Mamba (CVMamba): CVMamba utilizes the hierarchical feature representations produced by the VSS blocks to refine noisy labels progressively. By leveraging the selective scan capabilities of the SS2D module, the corrected labels align more accurately with the true underlying patterns in the data, thus enhancing the model's overall performance and resilience against label noise.
- Entropy-Divergence K-Nearest Neighbors Relabel (EDKR): EDKR corrects labels by combining Entropy-Divergence with KNN. It extracts features using a pre-trained network, calculates prediction probabilities, and uses divergence to assess confidence. Low-entropy, high-confidence samples are relabeled. KNN evaluates sample proximity, normalizes similarity scores, and identifies clean samples based on consistency for improved training.
- AdaptiveMix Data Augmentation: AdaptiveMix addresses class imbalances by dynamically adjusting instance counts to match the most populous class, ensuring balanced model training. Unlike traditional

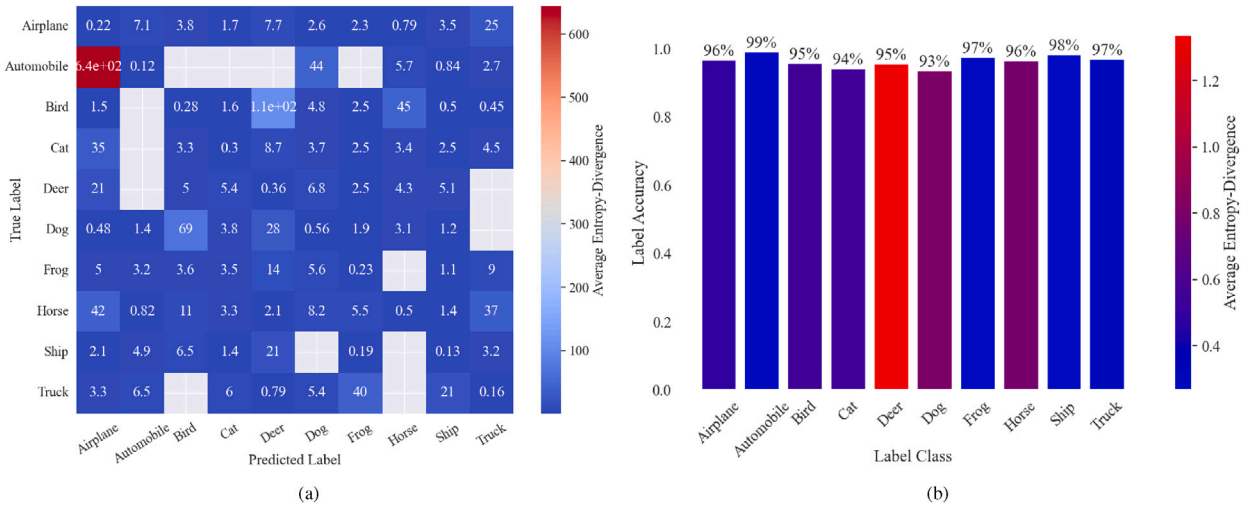


Fig. 1. On the left, a heatmap compares the average entropy-divergence values between predicted and true labels. Darker red shades indicate higher entropy values, with x-axis showing predicted categories and y-axis representing true categories. On the right, the post-correction accuracy by class is depicted. Darker blue bars indicate lower model uncertainty and lower entropy.

Mixup, AdaptiveMix refines the mixup coefficient from a Beta distribution, adjusting augmentation intensity during training. Our method enhances model generalization by effectively using both clean and noisy data, training the encoder with cross-entropy loss on mixed data.

2. Related works

2.1. State space models

As deep learning has advanced, various models have been developed to address noisy labels. Traditional approaches often use CNNs or ViTs. CNNs capture only immediate spatial relationships due to their local receptive fields, while ViTs and their variants can construct long-range dependencies with a global receptive field but face increased computational complexity with larger input sizes. The Structured State-Space Sequence Model (S4), introduced in [37], models long-range dependencies with linear complexity benefits. Subsequent models like S5 [38], H3 [39], and GSS [40] have built upon this. The Mamba model [18] incorporates a data-dependent SSMs layer and a selection mechanism (S6), excelling in large-scale datasets with linear scaling relative to sequence length.

It has been applied to various vision tasks, consistently achieving competitive outcomes. VMamba [19] extends the Mamba architecture to 2D image data, improving computer vision task performance by fully capturing interrelations among image patches. U-Mamba [41]integrates integrates SSMs into a U-Net architecture for medical image segmentation, capturing long-range dependencies and improving segmentation accuracy across different modalities. Swin-UMamba [42] merges Mamba's sequential capabilities with Swin Transformer's hierarchical structure to improve segmentation in 3D medical imaging. SegMamba [43] adapts the Mamba model for 3D segmentation tasks, leveraging state-space modeling for enhanced spatial learning in medical image processing. PyramidMamba [44] applies Mamba's state-space modeling to multi-scale feature fusion, optimizing semantic segmentation in remote sensing imagery. RS-Mamba [45] utilizes the state-space properties of Mamba for efficient classification of complex remote sensing images, enhancing the model's ability to handle diverse landscapes. TransMamba [46] integrates transformer architectures with Mamba's state-space modeling, leveraging the strengths of both for better performance in natural language processing tasks.

LocalMamba [21] adapts Mamba for vision tasks using a windowed selective scan mechanism, focusing on localized image patches while preserving local spatial relationships, making it effective for detailed spatial analysis. EfficientVMamba [47] introduces an atrous scanning technique to VMamba, reducing computations while maintaining the ability to model long-range dependencies, making it suitable for resource-constrained environments. SiMBA [48] simplifies Mamba for faster inference, employing a streamlined state-space representation that minimizes parameters, achieving a balance of speed and accuracy for real-time applications. MambaMorph [49] combines Mamba models with CNNs for accurate deformation field calculations, enhancing traditional CNN architectures for complex vision tasks. Vision Mamba [50] uses bidirectional SSMs and position embeddings to ensure precise visual recognition, efficiently processing large datasets and enhancing spatial interpretation in noisy environments.

In contrast, no prior work has systematically investigated the use of SSMs, or their visual adaptations, specifically against noisy labels within image classification tasks. To address LNL issues, we propose integrating Vision Mamba with CNNs in an attention-based fusion mechanism, effectively leveraging the benefits of SSMs. This hybrid model combines Vision Mamba's global context handling with CNNs' hierarchical, local feature-capturing abilities, managing vast data volumes without sacrificing speed or accuracy. LNL poses a unique challenge, with noisy and inaccurate data often hampering effective learning. The strength of our method, in extracting prominent features even from noisy data, demonstrates a promising path forward for advancing deep learning capabilities in the context of LNL during the warm-up stage.

In addition to this, one challenge with LNL is its potential to generate imprecise representations, which can impact downstream decision-making. This highlights the necessity of exploring the different categories of LNL algorithms, which can generally be divided into three solutions.

2.2. Correction of noisy labels

MentorNet [33] employs a student network by generating pseudo-labels using a pre-trained network. MetaCleaner [51] predicts the confidence scores of noisy samples based on their proximity in the feature space, which are then used to obtain cleaner representations. To deal with noisy labels, Yi and Wu [52] gradually adjust the data labels based on the predicted labels given by the network. The essence of the paper by Goldberger et al. [26] is encapsulated in the integration of noise adaptation layers utilizing noise transition matrices to enhance deep neural network training. Wang et al. [28] propose Symmetric Cross-Entropy

Learning, using Reverse Cross-Entropy to enhance training robustness and prevent overfitting. Han et al. [53] present a robust deep self-learning framework that effectively trains on noisy datasets without assumptions about label noise distribution or additional clean supervision. Xia et al. [54] propose TCC-Net, a two-stage method combining teacher–student networks for confidence correction. The teacher estimates label confidence, which the student uses for robust learning. However, its reliance on the teacher's accuracy makes it less effective when initial confidence estimates are unreliable, particularly in high noise situations. Zhang et al. [55] present UNICON + HMW, which combines unsupervised contrastive learning with Hard Mixup Weighting to enhance feature robustness. Although it handles label noise well, its complex training process and tuning needs limit scalability for large datasets. Chen et al. [56] present Robust LR*, a two-stage method that refines labels and then retrains the model. Despite its robustness, it requires precise calibration and may struggle with evolving noise patterns or insufficient initial refinements. Li et al. [57] introduce RML-Semi, a semi-supervised approach that iteratively refines mislabeled samples using both labeled and unlabeled data. Its effectiveness depends on accurate initial groupings, with early misclassifications potentially propagating errors. Zong et al. [58] propose DPC*, which uses Dirichlet Process Clustering to adaptively correct labels without a fixed number of clusters. Its flexibility is balanced by high computational costs and convergence challenges, making it less practical for large datasets. However, its performance is prone to deterioration under high noise rates and a large number of classes, often necessitating manual threshold adjustment for effective separation during different epochs of training.

2.3. Selection of confident pairs

In recent years, the focus on sample selection has gained increasing recognition in the LNL domain. A distinction between these methods lies in the criteria used for selecting clean samples. Early methodologies predominantly targeted the minimization of the impact of noisy samples by concentrating on training with cleaner subsets [30,33]. Current SOTA approaches lean towards incorporating semi-supervised learning paradigms [59]. Kim et al. [31] developed a method, SelNLPL, specifically aimed at addressing noisy labels in CNNs. A selection technique using prediction likelihoods for sample separation was proposed by Arpit et al. [60]. Song et al. [61] introduced TAkS, which selects reliable samples based on temporal consistency across training epochs. It assumes correctly labeled samples will show consistent predictions. This can be problematic if noisy labels remain stable over time, leading to their inclusion in training. Higashimoto et al. [62] developed CRAS, which uses contrastive representation learning with adaptive sample selection. This approach requires significant computational resources and may be less effective with complex or unknown noise distributions. Lin et al. [63] proposed DSCL+, which uses dynamic sample selection and contrastive learning for noise robustness. Its success depends on the accurate pacing of sample learning, which can be challenging with unreliable initial predictions or high noise levels. However, these methods can struggle in the face of intricate noise patterns.

In contrast, our proposed method handles various types of label noise with minimal to no alteration in the hyperparameter settings under different label-noise scenarios, including those with differing rates and types of noise. We demonstrate how a minimalistic approach to the selection process significantly enhances classification performance, surpassing SOTA methods in many instances.

2.4. Clean label usage

Earlier methods often reduce the influence of noisy samples by selecting a clean subset and training only with it [9,30,33]. Recent methods tend to further employ semi-supervised learning techniques, such as Mixup [34], MetaMixUp [35] and Mixup Inference [64], to fully explore the entire dataset by treating the selected clean subset as labeled samples and the non-selected subset as unlabeled samples [32,65]. CutMix [36] randomly cuts and pastes patches among training images, Puzzle Mix [66] rearranges parts of images in a puzzle-like fashion to create complex composites, and MixMatch [59] bridges the gap between labeled and unlabeled data through a unified loss function, blending labels and inputs for a more stable training process. AugMix [67] by Hendrycks et al. introduces a data augmentation strategy that improves model robustness and accuracy, especially under distributional shifts. IPMix approach by Huang et al. [68] combines multi-level augmentations to enhance precision and resilience in noisy settings without extra computational cost. Zhang et al. [69] propose Robust DivideMix, which improves noise modeling using Gaussian Mixture Models. While effective at handling higher noise rates, it depends on accurate noise estimation and struggles with non-random noise and severe class imbalance. Cordeiro et al. [70] introduce LongReMix, which combines long-tailed recognition with mixup strategies for noisy, imbalanced datasets. It requires careful parameter tuning and may underperform with extreme class imbalances or noise in majority classes. However, these methods typically use a static mixup strategy during training, which may not effectively address varying augmentation needs across different stages.

3. Methodology

Our methodology is outlined beginning with the establishment of relevant notations, followed by a detailed description of the proposed framework. We denote the set of labeled pairs as $(\mathcal{X}, \mathcal{Y}) = \{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\}$, where $\mathcal{X}$ represents the data and $\mathcal{Y}$ represents the corresponding labels. Here, each y_i is a one-hot encoded vector representing the class label out of C possible classes.

3.1. Convolutional VMamba (CVMamba)

3.1.1. Preliminaries

Our research primarily focuses on utilizing structured state space models (SSMs), including the structured state space sequence models (S4) and the Mamba model, for sequence data. These models are well-suited for processing sequences where the input $x(t) \in \mathbb{R}^H$ and the output $y(t) \in \mathbb{R}$ are mediated through a hidden state $h(t) \in \mathbb{R}^N$. The hidden state evolves according to an evolution matrix $A \in \mathbb{R}^{N\times N}$, with projection matrices $B \in \mathbb{R}^{N\times 1}$ and $C \in \mathbb{R}^{1\times N}$ defining the transformations:

$$h'(t) = Ah(t) + Bx(t), \quad y(t) = Ch(t). \tag{1}$$

For analyzing sequence data, we adapt these models to a discrete framework with a timescale parameter Δ, employing the zero-order hold (ZOH) method:

$$\bar{A} = \exp(\Delta A), \quad \bar{B} = (\Delta A)^{-1}(\exp(\Delta A) - I) \cdot \Delta B. \tag{2}$$

In the discrete model, the system equations transform to:

$$h_t = \bar{A}h_{t-1} + \bar{B}x_t, \quad y_t = Ch_t. \tag{3}$$

The output is computed through a global convolution, represented as:

$$K = (CB, CAB, \ldots, CA^{M-1}B), \quad y = x * K, \tag{4}$$

where M denotes the input sequence length, and K is a convolutional kernel.

3.1.2. Selective scan mechanism

To address the limitations of LTI SSMs Eq. (1) in capturing contextual information, Gu et al. [18] introduce an innovative parameterization method for SSMs, incorporating an input-dependent selection mechanism, known as S6. Despite this advancement, the selective SSMs encounter challenges in efficiently computing hidden states due to the time-varying weighting parameters, as convolutions cannot handle dynamic weights effectively. However, the recurrence relation of h_t in

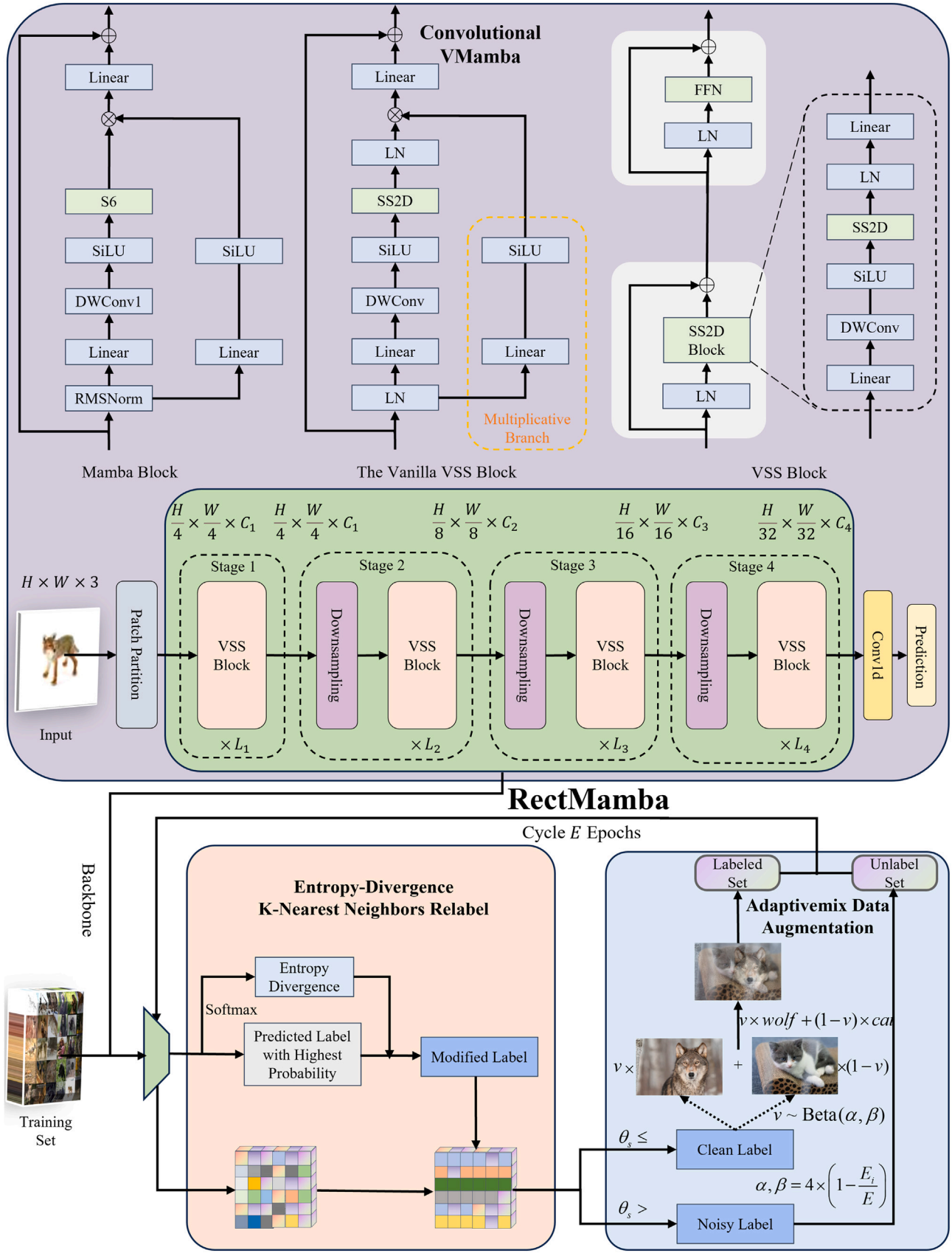


Fig. 2. Overview of RectMamba framework which includes: the CVMamba for tackling the warm-up phase during pre-training; the EDKR for noisy label correction; and the Adaptivemix method for adjustable data augmentation intensity throughout training epochs. Note: The Mamba block (left) shows the original S6-based design for comparison only. All actual vision components (Vanilla VSS Block, VSS Block, CVMamba) use SS2D instead of S6.

Algorithm 1 RectMamba with CVMamba Backbone and EDKR.

Require: Noisy labeled data $\mathcal{X}, \mathcal{Y}$, Pre-trained network f_θ, Total epochs E

for epoch $E_i = 1$ to E **do**:
  Divide input image $I \in \mathbb{R}^{H\times W\times 3}$ into patches.
  Apply $X \in \mathbb{R}^{H/4\times W/4\times C_1}$ to VSS blocks.
    ▷ Each VSS block uses SS2D modules (not S6) for 2D spatial processing.
  Softmax predictions $p(y_i|f_i)$ = Eq. (5)
  Compute entropy $H(p)$ = Eq. (7)

$$D(p) = h_{s1} - h_{s2}$$

  Update threshold θ_h by Eq. (6)
  if $H(p) < \theta_h$:
    $y_i^r \leftarrow l_i^r$
  else
    l_i^r is likely to be wrong.
  Apply KNN to select refined labels.
  Updated α, β by Eq. (11)
  $\upsilon = \max(\upsilon, 1-\upsilon)$
  x_{mix} = Eq. (12)
  y_{mix} = Eq. (13)
  Network forward propagation.
  Calculate the cross-entropy loss:

$$L_{ce} = -y_{mix}^T \log f(x_{mix})$$

  Classifier loss backpropagation.
end for
Output: Trained network f_θ and refined labels.

Eq. (3) can still be derived, allowing the response y_t to be computed efficiently using associative scan algorithms [71], which operate with linear complexity.

3.1.3. Network architecture

The architecture of CVMamba is outlined in Fig. 2, and we include pseudocode in Algorithm 1 to clarify the implementation and execution process. Initially, the input image $I \in \mathbb{R}^{H\times W\times 3}$ is divided into patches by a stem module, producing a 2D feature map with a spatial dimension of $H/4\times W/4$. Then, several network stages generate hierarchical representations with resolutions of $H/8\times W/8$, $H/16\times W/16$, and $H/32\times W/32$. Each stage includes a down-sampling layer (except for the first stage) followed by a series of Visual State Space (VSS) blocks.

VSS blocks act as the visual equivalent of Mamba blocks [18] for representation learning. The initial design of VSS blocks (termed the 'vanilla VSS Block') involves fully replacing the original S6 module from Mamba with the newly proposed 2D-Selective-Scan (SS2D) module. This substitution is necessary because S6 is designed for 1D sequential data, which is unsuitable for visual inputs that require 2D spatial modeling. SS2D preserves the dynamic selectivity and global receptive field of S6 while extending it to spatial data by processing patches along multiple 1D traversal paths and fusing them through cross-directional aggregation. Additionally, the multiplicative gating branch (highlighted in yellow) is removed, since the selectivity function of SS2D makes this component redundant. As a result, the VSS block contains a single processing branch with two residual modules, structurally similar to a transformer block [72].

3.1.4. 2D-selective-scan for vision data (SS2D)

While the sequential nature of the scanning operation in S6 is well-suited for NLP tasks that involve temporal data, it presents a notable challenge when applied to vision data, which is fundamentally non-sequential and incorporates spatial information (such as local texture and global structure). To tackle this challenge, S4ND [73] reformulates SSMs using convolutional operations, extending the kernel from 1D to 2D via the outer product. However, this adjustment makes the weights input-dependent, thus limiting the ability to capture contextual information effectively. To overcome this, we follow the selective scan approach [18] for input processing and introduce the 2D-Selective-Scan (SS2D) module. This adaptation allows S6 to be applied to vision data without losing its inherent benefits. Given the input data, SS2D unfolds the input patches into sequences along four unique traversal paths, where each sequence is processed with an input-dependent selectivity mechanism adapted from the principles of the S6 block. This adaptation ensures that the selectivity benefits of S6 are extended to 2D data while maintaining computational efficiency The resultant sequences are reshaped and merged to generate the output map (termed Cross-Merge). By employing complementary 1D traversal paths, SS2D allows each pixel in the image to integrate information from all other pixels in various directions, thus enabling the creation of global receptive fields within the 2D space.

3.2. Entropy-divergence k-nearest neighbors relabeling (EDKR)

As illustrated in Fig. 2, our proposed EDKR strategy combines an Entropy-Divergence label correction mechanism with a k-Nearest Neighbors (KNN) approach among training samples to rectify incorrect labels. We begin by employing a backbone network, a CVMamba pre-trained on ImageNet, to extract features and generate a global distance matrix encompassing all training samples. In this section, input data $\mathcal{X}$ is processed through an encoder f to extract feature vectors. For brevity, we define f_i as the representational feature of the sample x_i. Subsequently, a classifier processes these features to derive prediction scores for each class. These predictions are aggregated, and the features in the feature bank are normalized using L_2 normalization to ensure consistency in their magnitudes.

During the sample re-labeling stage, softmax activation is applied to the aggregated prediction scores, yielding a normalized prediction matrix, which can be represented as:

$$p(y_i|f_i) = \frac{e^{z_i}}{\sum_{j=1}^{C} e^{z_j}}, \tag{5}$$

where $p(y_i|f_i)$ is the probability of label y_i given feature vector f_i, and z_i is the logit corresponding to class i.

To ensure that only samples with high confidence are relabeled, our threshold θ_h integrates three key components: entropy $H(p)$, divergence $D(p)$, and an epoch-adaptive scaling factor $\frac{E_i}{E}$. This design balances uncertainty and confidence, adapting over the course of training. We sort the entropy values and define a threshold θ_h based on the epoch-wise proportion of the dataset:

$$\theta_h = \frac{H(p)}{D(P)} \times \frac{E_i}{E}. \tag{6}$$

Here, E_i is the current epoch and E is the total number of epochs. Samples with entropy below θ_h are considered highly confident. Their original labels are replaced by the model's predicted labels h_{l1}.

1. Entropy $H(p)$: Entropy calculations are then performed to gauge the model's prediction uncertainty for each sample, formulated as:

$$H(p) = -\sum_{i=1}^{C} p(y_i|f_i)\log(p(y_i|f_i)). \tag{7}$$

Eq. (7) is based on the Shannon entropy [74], which measures the uncertainty of a probability distribution. In our context, $H(p)$ quantifies the uncertainty in the predicted class probabilities and is a proxy for the model's confidence in its predictions. This metric has been used in deep learning tasks, including confidence estimation and semi-supervised

learning. In particular, Grandvalet and Bengio [75] were among the first to incorporate entropy into neural network training, proposing entropy minimization to encourage confident model outputs. Inspired by this, we leverage entropy in Eq. (7) to dynamically adjust the threshold for label correction.

By incorporating entropy into our threshold, we ensure that samples with higher uncertainty (i.e., higher entropy values) have a stricter threshold, reducing the likelihood of incorrect label correction. This prevents erroneous relabeling when the model is uncertain, which could otherwise amplify noise in the data. For example, consider the predicted probabilities shown in Fig. 3, where multiple class probabilities are similar, resulting in a high entropy value of 3.3205. The model's confidence in any particular prediction is low, and according to our thresholding strategy, this sample would have a stricter threshold, making it less likely to be relabeled. This avoids propagating errors in the training process.

In contrast, the predicted probabilities in Fig. 4 are more concentrated on a single class, resulting in a lower entropy value of 1.0656. The model is more confident in its prediction, indicating that it is highly likely that the predicted label (*Horse* in this case) is correct. As a result, the threshold becomes less restrictive, allowing confident samples like this one to be relabeled if necessary.

2. Divergence $D(p)$: The model's highest and second highest confidence predictions and their associated labels are extracted as $h_{s1}, h_{l1} = \max_{i1}(p(y_i|f_i))$ and $h_{s2}, h_{l2} = \max_{i2}(p(y_i|f_i))$, respectively. Divergence $D(P) = h_{s1} - h_{s2}$ quantifies the confidence of a prediction by measuring the difference between the highest and second-highest predicted class probabilities. A larger divergence indicates higher confidence in the prediction. This approach is inspired by the Distinguishable

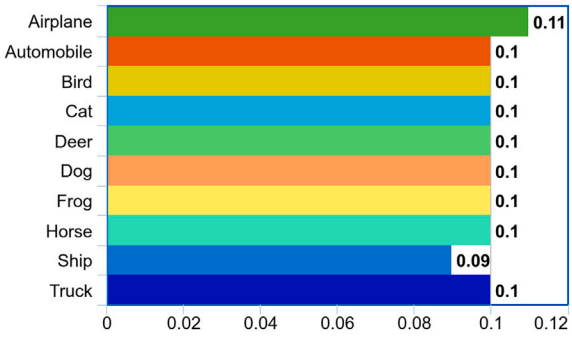


Fig. 3. This image displays multiple class prediction probabilities, with no single class dominating. This results in a high entropy value (3.3205), indicating that the model is unsure about the correct label.

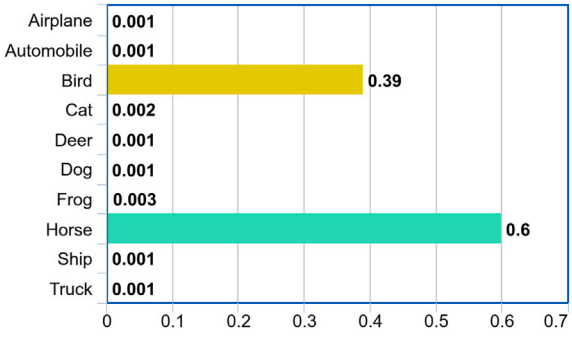


Fig. 4. This image shows a clear peak in predicted probability for one class, resulting in low entropy (1.0656), which signifies high confidence in the prediction. The model predicts the class 'Horse' with a high level of certainty. Although the model shows certainty, the close values between the top two predictions indicate a moderate confidence level in the prediction.

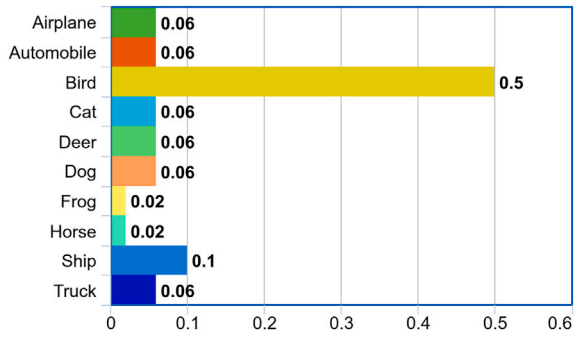


Fig. 5. This image displays predicted probability values with a higher entropy (2.5191), yet the clear separation between the top two predicted probabilities (Bird category) signifies that the model has a high level of confidence in this prediction.

Probability-Aware Distribution introduced in [76], where the difference between the top predicted probabilities is used to separate noisy labels. When $D(p)$ is large, it not only indicates that h_{s1} is much greater than h_{s2} but also implies that the sum of probabilities for all other classes is relatively smaller. This concentration of probability mass demonstrates a strong belief in the correctness of the prediction.

For instance, in Fig. 4, although the entropy is low, the difference between the highest and second-highest predicted probabilities is not particularly large. This suggests that, despite the low entropy, the model's confidence in selecting the true label over the second choice is moderate. In contrast, the predictions in Fig. 5 exhibit a higher entropy value of 2.5191, indicating more uncertainty overall. However, the divergence $D(p)$ is significantly larger, suggesting that the model is more confident about the highest predicted label despite the higher entropy. This emphasizes the importance of considering both entropy and divergence to accurately assess prediction confidence.

3. Epoch-Adaptive Scaling $\frac{E_i}{E}$: The ratio $\frac{E_i}{E}$ is an epoch-based scaling factor that adjusts the threshold as training progresses. At earlier epochs, when E_i is smaller relative to E, the threshold θ_h is higher, conservatively limiting label correction while the model is still learning and potentially less accurate. As training continues, this factor increases, lowering the threshold and enabling more aggressive label correction as the model becomes more confident.

By combining these components, the term $\frac{H(p)}{D(p)}$ effectively balances uncertainty and confidence. There is an inverse relationship between $H(p)$ and $D(p)$: when $H(p)$ is high (greater uncertainty), $D(p)$ tends to be small (lower confidence), and vice versa. By integrating both, we ensure that only samples for which the model is both confident and certain are selected for label correction.

In the KNN (k_q) sample selection phase, the feature space is utilized to evaluate sample proximity, where $k_{q_i} \triangleq k_q(f_i)$ denotes the prediction vector. The utilization of KNN for selecting clean samples is grounded in the principle of *local consistency* in the feature space [77]. Samples that are close to each other are likely to share the same label. In datasets with noisy labels, mislabeled samples tend to have features that are inconsistent with their assigned labels. By examining the immediate neighborhood of a sample, we can assess the consistency of its label with those of its nearest neighbors. KNN effectively aggregates information from multiple samples, reducing the influence of noise in any single label. This aggregation helps distinguish correctly labeled samples from mislabeled ones based on the agreement among neighbors. Compared to other methods, KNN is computationally efficient and scales well with large datasets and high-dimensional feature spaces, making it suitable for practical applications in deep learning contexts.

Each sample feature is compared with every other feature in the feature bank to measure similarities. The affinity between the representations f_i and f_j of samples x_i and x_j is expressed as s_{ij}. This

can be mathematically represented using the cosine similarity as:

$$s_{ij} = \frac{f_i^T f_j}{\|f_i\|_2 \|f_j\|_2} \tag{8}$$

To accentuate the influence of top-k nearest features, weights are determined based on their similarity values. Before the assignment of weights, the similarity scores are normalized: $min_i = \min(s_{ij})$, $max_i = \max(s_{ij})$, where i and j iterate over the top-k nearest neighbors.

$$S_{ij} = \frac{s_{ij} - min_i}{max_i - min_i}. \tag{9}$$

Given the weighted similarities, a score matrix is generated, reflecting the harmony between KNN inferred predictions and the modified label. A predefined threshold distinguishes clean samples. Those surpassing this threshold, suggesting significant concordance among distinct prediction strategies, are recognized as *clean*.

Consistency for each sample is calculated using a weighted sum of its nearest neighbors' labels with the normalized weights, defined as:

$$c_i = \sum_{j=1}^{k} S_{ij} \times y_j. \tag{10}$$

where y_j is the one-hot encoded label for the j-th nearest neighbor. Samples with c_i above a certain threshold are included in the clean subset $(\mathcal{X}_c, \mathcal{Y}_c)$.

By integrating the entropy-divergence measures with KNN-based neighborhood information, our method leverages both the model's prediction confidence and the structural relationships among samples. This combination enhances the robustness of label correction, effectively identifying and rectifying mislabeled samples while preserving correctly labeled ones.

3.3. Adaptivemix data augmentation

Our methodology adeptly addresses class imbalances within datasets by dynamically calibrating the number of instances across each class to align with that of the most populous one. This balanced approach is crucial in classification tasks, particularly when dealing with underrepresented classes. It facilitates unbiased model training and fosters uniform generalization across all categories.

Studies have shown that the effectiveness of data augmentation techniques can vary at different training stages. Zhang et al. [34] introduced Mixup, demonstrating that it improves model robustness by generating convex combinations of training examples. Guo et al. [78] noted that the benefits of Mixup depend on proper tuning of the mixing parameters and that over-augmentation can harm performance. By adapting the mixing coefficients, we address these concerns:

- Preventing underfitting: Avoiding overly aggressive augmentation in later stages ensures that the model does not underfit the data.
- Improving convergence: Gradually reducing augmentation intensity aids convergence by allowing the model to focus on minimizing loss on actual data samples.

To overcome this, we introduce the AdaptiveMix approach, which dynamically adjusts the intensity of data augmentation. Our method aims to:

- Enhance generalization in the early stages by encouraging the model to explore a broader data manifold.
- Allow fine-tuning and convergence in later stages by reducing the augmentation intensity, enabling the model to focus on the actual data distribution.

The mixing coefficients α and β in our method are defined as follows:

$$\alpha, \beta = 4 \times \left(1 - \frac{E_i}{E}\right), \tag{11}$$

where E_i is the current epoch number, and E is the total number of epochs. The Beta distribution is a flexible family of continuous probability distributions defined on the interval $[0, 1]$, parameterized by α and β. Its shape can model various behaviors:

- When $\alpha = \beta > 1$: The distribution is unimodal and symmetric around 0.5, concentrating probability mass near the center.
- When $\alpha = \beta = 1$: The distribution becomes uniform over $[0, 1]$.
- When $\alpha = \beta < 1$: The distribution is U-shaped, with more mass near 0 and 1.

By setting $\alpha = \beta$ and dynamically adjusting them over time, we control the shape of the Beta distribution from which the mixing coefficient υ is sampled. Here, υ is derived from a *Beta* distribution parameterized by α and β: $\upsilon \sim \text{Beta}(\alpha, \beta)$. To ensure υ remains within a balanced range, it is refined as: $\upsilon = \max(\upsilon, 1 - \upsilon)$. In the mixup context, υ is used to linearly combine two inputs and their labels. For instance, for two inputs x_a and x_b with their associated targets y_a and y_b, the mixed input and target are derived as:

$$x_{mix} = \upsilon \times x_a + (1 - \upsilon) \times x_b. \tag{12}$$

$$y_{mix} = \upsilon \times y_a + (1 - \upsilon) \times y_b. \tag{13}$$

At the start of training ($E_i = 0$), $\alpha = \beta = 4$, resulting in a Beta distribution peaked around 0.5, leading to strong mixing where inputs are mixed equally. This exposes the model to more challenging, augmented data, promoting robustness and preventing early overfitting. High augmentation encourages the model to learn generalized features.

As training progresses ($E_i \to E$), $\alpha, \beta \to 0$. The distribution becomes U-shaped, favoring values closer to 0 and 1. Since we take $\upsilon = \max(\upsilon, 1 - \upsilon)$, υ is confined to $[0.5, 1]$, thus preferring values closer to 1 in later epochs, resulting in less mixing. This allows the model to focus on the actual data distribution, refining its understanding and improving performance on the true data. Reduced augmentation enables the model to fine-tune these features to the specific data.

This blended data, generated using the mixup coefficient υ, is then incorporated into the training process. For each sample pair x, y_{mix} in the selected subset $(\mathcal{X}, \mathcal{Y}_{mix})$, the encoder f is trained using a common cross-entropy loss, formulated as: $L_{ce} = -\boldsymbol{y_{mix}}^T \log f(\boldsymbol{x})$. Adaptivemix is grounded in a strategic approach to data augmentation that:

- Dynamically adjusts augmentation intensity to match the model's training needs at different stages.
- Leverages theoretical properties of the Beta distribution to control the mixing coefficient.
- Incorporates principles from curriculum learning to enhance model performance and convergence.

4. Experiments

This paper presents a comprehensive evaluation of our proposed method, encompassing both quantitative and qualitative analyses across four datasets. The experimental setup, including dataset descriptions and parameter configurations, is detailed in Sections 4.1 and 4.2. The ablation study in Section 4.3 highlights the effectiveness of our model's core components: entropy-divergence and KNN label rectification, adaptive mixup dataset enhancement, and convolutional vision Mamba pretrained backbone. Section 4.4 offers a theoretical perspective on how SSMs inherently suppress label noise through recurrence-driven smoothing, selective filtering, and temporal regularization. Additionally, Section 4.5 provides a robustness analysis of the KNN parameter k, further demonstrating the flexibility of our model.

4.1. Datasets

We evaluate our method using the CIFAR-10 and CIFAR-100 datasets, introduced by Krizhevsky et al. [79], which feature 60,000 images in total, divided into distinct classes. The CIFAR-10 dataset comprises 10 classes, while CIFAR-100 includes 100 classes grouped into 20

superclasses. To assess noise resilience, we employ synthetic noise models, both symmetric and asymmetric. The asymmetric noise simulates real-world mislabeling errors, particularly in CIFAR-100, where label shifts depend on the superclass.

The Clothing1M dataset, developed by Xiao et al. [25], consists of 1 million noisy images across 14 clothing categories sourced from online shopping websites. Labels are derived from seller descriptions, resulting in a high mislabeling rate, especially among similar categories like *Knitwear* and *Sweater*. The noise ratio is estimated to be around 38.5 %. We also include the ANIMAL-10N dataset, a recent smaller real-world dataset with 10 animal classes, proposed by Song et al. [80]. It contains 50,000 training and 10,000 testing images, with an estimated 8 % error rate due to manual labeling errors.

4.2. Training details

The CVMamba network, used for the CIFAR-10/100 and ANIMAL-10N datasets, starts with weights pre-trained on ImageNet. To ensure fair comparisons with existing methods that do not utilize pre-trained models, we also conducted experiments using a ResNet18 backbone without pre-training for the CIFAR-10 and CIFAR-100 datasets. The ResNet18 model was trained from scratch using the same training settings as the CVMamba network, except that the initial weights were randomly initialized. The model is trained using a Stochastic Gradient Descent (SGD) optimizer, with a learning rate (τ) set at 0.0005 and a momentum parameter of 0.9. The training proceeds with a batch size of 20 over a total of 50 epochs. This allows us to directly compare the effectiveness of our method with and without pre-training.

For the Clothing1M dataset, we employ the ResNet50 [1] model with ImageNet pretrained weights as followed in the work by Li et al. [32]. We train the network with SGD optimizer for 150 epochs with a momentum of 0.9 and weight decay of 0.001. The initial learning rate is set to 0.002 and is reduced by a factor of 10 at the 50th and 100th epochs, with a fixed batch size of 32.

4.3. Ablation study

We evaluate our noisy label correction method on four datasets with uniform settings: 50 % symmetric noise, 50 epochs, using the CVMamba pre-trained backbone. The experiments are categorized into five groups, each demonstrating a core contribution of our model:

- ResNet18: This group evaluates the ResNet18 model's performance by training it directly on a noisy dataset without label correction mechanisms.
- CVMamba: By comparing the results to those of ResNet18, we can isolate and demonstrate the unique contributions and improvements provided by CVMamba.
- SSR+: The baseline method, utilizing a SOTA correction approach.
- CVMamba+EDKR: An enhancement of the CVMamba model, this variant integrates the Entropy-Divergence K-Nearest Neighbors Relabel (EDKR) method to correct the noisy labels.
- RectMamba: This variant combines Mamba label rectification with AdaptiveMix. The addition aims to enhance the model's robustness during the warm-up stage, thereby achieving superior label correction performance.
- RectMamba+LP: To isolate the effect of the EDKR module, we replace it with classical Label Propagation (LP) [81], keeping all other components of RectMamba unchanged.

Initially, our focus is on validating the benefits of Mamba in LNL. Table 1 indicates that CVMamba significantly outperforms ResNet18 in terms of test accuracy. Specifically, CVMamba achieves a test accuracy of 86.8 %, compared to ResNet18's 75.2 %. This improvement highlights the unique contributions and enhancements introduced by the CVMamba model. ResNet18, when trained directly on a noisy dataset without any label correction mechanisms, shows a baseline performance that is considerably lower. The superior performance of CVMamba suggests that it incorporates effective strategies to handle noisy data, leading to more accurate predictions.

Table 1
Performance comparison of methods in terms of Test Accuracy and True Positives (TP).

Method	Test accuracy	TP
ResNet18	75.2	–
CVMamba	86.8	–
SSR+	96.1	30,600
CVMamba+EDKR	96.6	47,865
RectMamba+LP [81]	95.8	33,544
RectMamba	**96.8**	**48,633**

Despite its known performance, SSR+ struggles with symmetric noise, with fluctuating correct label rates over 300 epochs indicating vulnerability to noise. EDKR stands out for its consistent and resilient performance, with a steadily ascending trajectory that highlights its ability to enhance label accuracy over time. Importantly, Entropy-Divergence functions without any parameters, streamlining its implementation and maintaining efficiency. Building upon CVMamba, the CVMamba+EDKR model, fortified with the KNN method for clean sample selection, pushed the boundaries even further. The results spotlight an uptick in both Test Accuracy and TP, registering figures of 96.6 % and 47,865 respectively. This underlines the efficacy of enhancing the label selection mechanism. These results strongly corroborate that harnessing Entropy-Divergence and KNN in the label correction paradigm is instrumental in gauging label certainty and facilitating more precise adjustments, thus emphasizing the pivotal role entropy can play in label correction.

To directly evaluate local consistency under noisy conditions, we introduced a classic Label Propagation (LP) baseline by replacing EDKR in RectMamba with the LP module from Zhou et al. [81]. A KNN graph was constructed using L_2-normalized CVMamba features from ImageNet-pretrained weights, and label diffusion was performed using cosine similarity. Under 50 % symmetric noise on CIFAR-10, LP achieved a test accuracy of 95.8 % and 33,544 true positives. While LP captures feature similarity effectively, it lacks uncertainty modeling, leading to occasional over-smoothing and incorrect propagation across class boundaries. Compared to classical LP (95.8 % accuracy, 33,544 TP), our proposed EDKR achieves higher accuracy (96.8 %) and identifies significantly more clean samples (48,633 TP). This demonstrates that EDKR not only preserves local consistency through KNN-based similarity, but also surpasses LP by integrating entropy-divergence filtering, which effectively distinguishes trustworthy predictions from mislabeled data and avoids over-smoothing across class boundaries.

Fig. 6 portrays a comparative analysis between our method, RectMamba, and the widely recognized SSR+. The horizontal axis

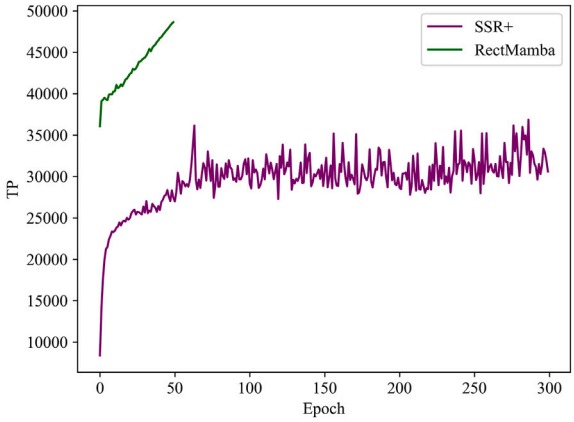


Fig. 6. Comparative analysis of label refine performance between RectMamba over 50 epochs and SSR+ over 300 epochs.

delineates the progression of epochs, while the vertical axis represents the TP after each label selection, which are eventually supplied for backbone training. It's evident that our RectMamba approach has exhibited a significant increase in the TP count by the end of the 50 epochs, achieving a TP of 48,633. In contrast, SSR + reaches a plateau earlier, recording a TP count of 30,600. The consistent upward trajectory of RectMamba demonstrates its capacity to effectively discern and select more accurate labels as epochs progress. The erratic behavior of SSR + , on the other hand, suggests its struggles with the challenges of label selection, especially in the latter epochs. This evidence supports that RectMamba, with its superior selection mechanism, harnesses the underlying data more efficiently than SSR + , ensuring more robust training for the backbone.

4.4. How SSMs suppress label noise

SSMs, as employed in CVMamba, contribute to robustness against label noise through three main mechanisms: (1) global receptive field smoothing via structured recurrence, (2) selective filtering of discriminative features, and (3) implicit regularization induced by temporal/spatial recurrence. We now provide a formal analysis of these effects.

In CVMamba, each VSS block updates hidden states h_t using the discrete-time SSMs formulation:

$$h_t = \bar{A}h_{t-1} + \bar{B}x_t, y_t = Ch_t. \tag{14}$$

Here, $\bar{A}$ acts as a temporal convolution operator with memory. When training on noisy labels $\tilde{y}$, small fluctuations (e.g., incorrect gradient signals from $\tilde{y} \neq y$) are smoothed over time by the recurrence. Since $\bar{A}$ is a stable matrix (eigenvalues inside the unit circle), the recurrence suppresses high-frequency (i.e., noisy) components of the gradient signal. Thus, the network exhibits low-pass filtering behavior:

$$\tilde{h}_t = \sum_{k=0}^{t} \bar{A}^{t-k}\bar{B}x_k \rightarrow \text{smooth integration of past inputs.}$$

The SS2D module processes feature sequences with input-dependent gating weights g_t, applied before recurrence:

$$\tilde{x}_t = g_t \cdot x_t, \quad h_t = \bar{A}h_{t-1} + \bar{B}\tilde{x}_t.$$

This modulated input $\tilde{x}_t$ effectively reduces the contribution of uncertain or uninformative features, often correlated with mislabeled samples, by assigning lower g_t values based on confidence. Since noisy gradients typically correspond to low-confidence activations, selectivity attenuates their propagation.

Unlike CNNs or Transformers that update weights independently per layer, SSMs enforce consistency through state transitions. This induces a form of temporal regularization. Consider the label prediction loss with respect to noisy label $\tilde{y}_t$:

$$L = \sum_{t=1}^{T} \ell(Ch_t, \tilde{y}_t), \quad \text{subject to } h_t = \bar{A}h_{t-1} + \bar{B}x_t.$$

Here, h_t is constrained by previous inputs and states. Therefore, even if $\tilde{y}_t$ is corrupted, the optimization does not freely overfit it because h_t cannot diverge sharply without violating recurrence consistency. This coupling of hidden states across time or spatial axes makes the model inherently more robust to local label noise.

4.5. Robustness analysis of the k-nearest neighbors parameter k

In this section, we investigate the impact of varying the number of nearest neighbors k on the classification accuracy of our proposed method under different noise conditions. The objective is to assess the robustness of the k parameter in the K-Nearest Neighbors (KNN) algorithm when applied to noisy datasets.

Table 2 presents the classification accuracies achieved on the CIFAR-10 and CIFAR-100 datasets with symmetric and asymmetric noise types, respectively, at specified noise rates. We vary the value of k from 50 to 500 to observe its effect on performance. It is evident that the classification accuracy remains relatively stable across different values of k. Specifically, for CIFAR-10 with 50 % symmetric noise, the accuracy fluctuates marginally between 96.8 % and 97.0 % as k increases from 50 to 500. Similarly, for CIFAR-100 with 30 % asymmetric noise, the accuracy varies slightly between 82.4 % and 83.5 %.

Table 2
Effect of varying the number of nearest neighbors k on classification accuracy (%) for CIFAR-10 and CIFAR-100 datasets under different noise conditions.

Dataset	Type	Rate	k	Accuracy (%)
CIFAR-10	Sym	50 %	50	97.0
			100	96.9
			200	96.8
			300	96.9
			500	96.9
CIFAR-100	Asym	30 %	50	83.5
			100	82.9
			200	82.6
			300	82.9
			500	82.4

These observations indicate that the performance of our method is not significantly sensitive to the choice of k within the tested range. The robustness can be attributed to the fact that the KNN algorithm, when integrated with our EDKR strategy, effectively identifies and corrects noisy labels even when the number of neighbors varies. The minor fluctuations in accuracy suggest that our method can maintain high performance without requiring precise tuning of the k parameter. This robustness is particularly advantageous in practical applications where the optimal value of k may not be known in advance.

In addition to its influence on accuracy and computational cost, the choice of k significantly impacts training stability. We observed that small k values (such as 50) led to unstable training behavior. This is because label correction becomes highly sensitive to local noise, resulting in oscillating performance and slower convergence. Conversely, large k values (such as 300–500) introduce too many weakly related or noisy neighbors, which can dilute correction quality and lead to oversmoothing, ultimately slowing down the learning process. The most stable performance was achieved with moderate k values (between 100 and 200), where the model demonstrated both fast convergence and low variance in validation accuracy. This range achieves a balance between local specificity and global robustness in the feature space, allowing label corrections to be both effective and reliable.

5. Results and comparisons

In this section, we present our RectMamba's performance under various label noise conditions, starting with synthetic datasets such as CIFAR-10/100 and progressing to real-world datasets like ANIMAL-10N and Clothing1M.

5.1. Symmetric noise

In the symmetric noise model, each label in the dataset is independently replaced with a different label at a fixed probability. This probability defines the noise rate, and the process is uniformly applied across all classes. For instance, if a symmetric noise rate of 20 % is applied, each label has a 20 % chance of being randomly assigned to one of the other classes, making this noise evenly distributed across all classes. As a result, every class label has the same probability of being corrupted, simulating an artificial scenario where labels are incorrectly assigned with no preference toward any particular class. This noise model is commonly used to evaluate the robustness of models against uniformly distributed errors.

Table 3
Test accuracies (%) obtained by different techniques under symmetric and asymmetric noise on CIFAR-10 and CIFAR-100 datasets. The best results are in bold. The data are copied from respective papers. The - denotes the absence of respective data.

Method	Symmetric Noise						Asymmetric Noise					
	CIFAR-10			CIFAR-100			CIFAR-10			CIFAR-100		
	20 %	50 %	80 %	20 %	50 %	80 %	10 %	30 %	40 %	10 %	30 %	40 %
CE	86.8	79.4	62.9	62.0	46.7	19.9	88.8	81.7	76.1	68.1	53.3	44.5
LDMI [82]	88.3	81.2	43.7	58.8	51.8	27.9	91.1	91.2	84.0	68.1	54.1	46.2
TCC-net [54]	91.9	89.3	–	68.9	62.4	–	–	89.2	82.7	–	62.9	54.6
TAkS [61]	83.9	76.8	–	50.7	41.0	–	–	–	73.4	–	–	35.2
PCIL [52]	92.4	89.1	77.5	69.4	57.5	31.1	93.1	92.9	91.6	76.0	59.3	48.3
JPL [83]	93.5	90.2	35.7	70.9	67.7	17.8	92.0	92.5	90.7	72.0	68.1	
UNICON + HMW [55]	93.5	95.2	93.7	76.6	75.8	63.4	93.5	94.7	93.7	76.7	76.3	72.1
MOIT [84]	94.1	91.1	75.8	75.9	70.1	51.4	94.2	94.1	93.2	77.4	75.1	74.0
Robust DivideMix [69]	95.5	–	61.6	77.4	–	48.1	–	–	86.9	–	–	–
M-Up [34]	95.6	87.1	71.6	67.8	57.3	30.8	93.3	83.3	77.7	72.4	57.6	48.1
CRAS [62]	95.7	95.4	93.8	79.5	77.0	65.0	–	–	93.1	–	–	78.0
RRL [85]	95.8	94.3	92.4	79.1	74.8	57.7	–	–	91.9	–	–	–
ELR + * [86]	95.8	94.8	93.3	77.6	73.6	60.8	95.4	94.7	93.0	77.3	74.6	73.2
NGC [87]	95.9	94.5	91.6	79.3	75.9	62.7	–	–	90.6	–	–	–
TCL + * [88]	96.0	94.5	–	79.3	74.6	–	95.9	94.8	93.7	79.0	76.9	–
UniCon* [89]	96.0	95.6	93.9	78.9	77.6	63.9	95.3	94.8	94.1	78.2	75.6	74.8
OT-Filter [90]	96.0	95.3	94.0	76.7	73.8	61.8	–	–	95.1	–	–	76.5
DivideMix* [32]	96.1	94.6	93.2	77.3	74.6	60.2	93.8	92.8	91.7	71.6	69.5	55.1
LongReMix [70]	96.3	95.1	–	77.9	75.5	–	–	–	94.7	–	–	59.8
LNL-FG [91]	–	95.2	–	–	74.7	–	–	–	94.9	–	–	75.1
DMLP-DivideMix [92]	96.3	95.8	94.5	79.9	76.8	68.6	–	–	95.0	–	–	–
AugDesc* [93]	96.3	95.4	93.8	79.5	77.2	66.4	–	–	94.6	–	–	–
DSCL + [63]	96.3	95.5	93.8	79.5	76.7	70.3	96.8	95.2	92.9	80.1	77.9	76.1
C2D* [94]	96.4	95.3	94.4	78.7	76.4	67.8	–	–	93.5	–	–	–
Robust LR* [56]	96.5	95.8	94.3	79.1	75.3	66.7	–	–	94.4	–	–	–
RML-Semi [57]	96.5	95.7	93.9	78.9	77.8	64.1	–	–	–	–	–	–
DPC* [58]	96.5	95.9	94.8	81.0	78.5	66.4	95.4	95.3	94.9	80.5	79.7	75.6
InstanceGM [95]	96.7	95.9	–	79.7	77.2	–	–	–	–	–	–	–
SSR + [96]	96.7	96.1	95.6	79.7	77.2	–	–	–	95.5	–	–	–
RegCE + semi [97]	96.8	96.0	93.5	80.5	77.3	66.4	–	–	92.2	–	–	66.8
RectMamba (ResNet18)	96.8	96.6	95.8	81.1	79.4	69.8	96.5	96.0	95.8	81.2	80.1	78.6
RectMamba (ViTs)	96.9	96.7	95.9	82.6	80.3	72.1	96.8	96.4	96.0	82.3	81.5	79.2
RectMamba (CVMamba)	**97.1**	**96.8**	**96.1**	**83.2**	**82.9**	**77.8**	**97.0**	**96.7**	**96.5**	**83.5**	**82.6**	**82.1**

We evaluated the RectMamba method under different levels of symmetric noise at 20 %, 50 % and 80 % in the CIFAR-10 and CIFAR-100 datasets, demonstrating significant improvements over existing techniques. Table 3 offers a comprehensive overview of several SOTA methods, ranging from conventional approaches like CE, LDMI [82], and PCIL [52] to more recent advancements such as DPC* [58], RML-Semi [57] and DSCL + [63]. Overall, it's apparent that the majority of techniques undergo a decrease in accuracy as the noise rate amplifies from 20 % to 80 %. This highlights the inherent challenges of higher noise rates and the subsequent difficulty in preserving model robustness under such adverse conditions. Our proposed RectMamba method distinguishes itself with exceptional accuracy and resistance to increasing noise levels. For example, in CIFAR-10, RectMamba attains test accuracies of 97.1 %, 96.8 % and 96.1 % at 20 %, 50 % and 80 % noise rates, respectively. Similarly, for CIFAR-100, the results are commendable at 83.2 %, 82.9 % and 77.8 %.

5.2. Asymmetric noise

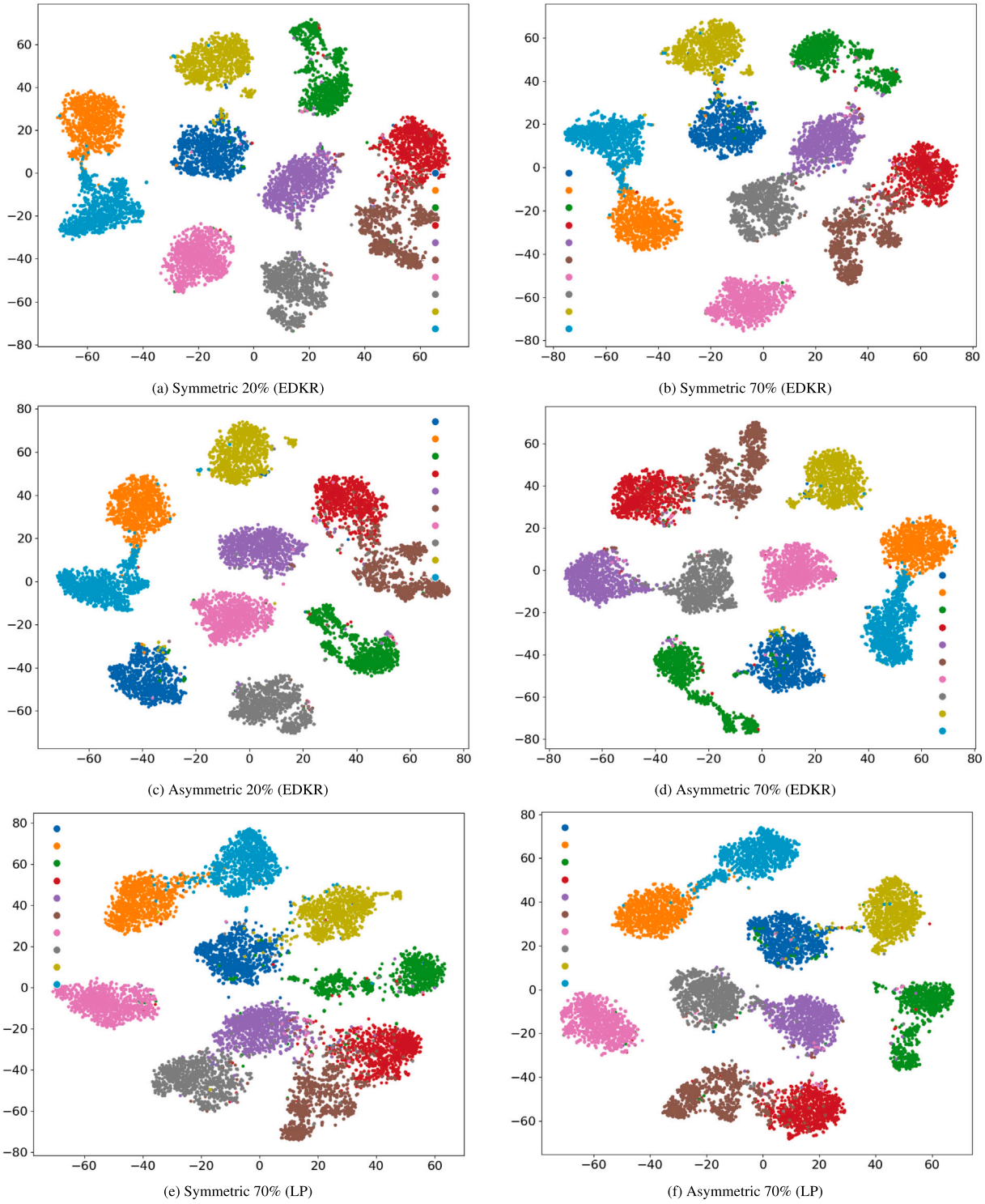
Asymmetric noise, on the other hand, is designed to reflect more realistic, class-specific mislabeling patterns that are frequently encountered in real-world datasets. In this model, certain classes are more prone to being mislabeled as similar classes. For example, in CIFAR-10, cats may be misclassified as dogs, and trucks may be mislabeled as automobiles. Similarly, in CIFAR-100, which contains 100 classes grouped into 20 superclasses, the noise is introduced in a way that label shifts occur within the same superclass. This means that a class label is more likely to be replaced by another label within the same group, which mimics errors that occur due to the intrinsic similarities between certain categories.

Further analysis of results for asymmetric noise in Table 3 confirms the effectiveness of our newly introduced RectMamba method. This table spans a variety of algorithms across two distinct asymmetric noise rates: 10 %, 30 % and 40 %. Remarkably, our proposed RectMamba method consistently achieves the highest test accuracies across both datasets and throughout all noise rates. For instance, when operating under a 10 % noise rate, it reaches 97.0 % and 83.5 % accuracies on CIFAR-10 and CIFAR-100, respectively, surpassing all other compared algorithms in LNL. The displayed results clearly attest to the robustness and adaptability of RectMamba in scenarios with asymmetric noise. This consistent performance confirms RectMamba as a viable solution for real-world applications facing significant label noise challenges.

Additionally, to address the concern of unfair comparisons due to the use of pre-trained models, we evaluated our method using a non-pretrained ResNet18 backbone, denoted as **RectMamba (ResNet18)**. The training settings for this experiment were consistent with those used for the CVMamba network, except that the ResNet18 model was trained from scratch without pre-trained weights. As shown in Table 3, **RectMamba (ResNet18)** achieves a test accuracy of 78.6 % on CIFAR-100 with a 40 % asymmetric noise rate, which is competitive with SOTA methods under the same conditions. This demonstrates that the performance gains of our method are not solely due to the use of a pre-trained backbone but are primarily attributed to our proposed label correction mechanism.

To ensure a comprehensive comparison, we further integrated the ViTs [98] backbone into our RectMamba framework, aiming to assess the performance of CVMamba. Our CVMamba contains approximately 33 M parameters, which lies between ResNet-18 (11.7 M) and ViTs-B (85.8 M). This parameter count illustrates that CVMamba achieves a

good balance between model capacity and efficiency—more expressive than ResNet-18 but significantly lighter than ViTs-B. Compared to ViTs's quadratic complexity in token length, state space models like Mamba enable linear-time sequence processing. We benchmarked average training time per epoch using the same hardware, where CVMamba requires approximately 43 s per epoch, compared to 19 s for ResNet-18 and 82 s for ViTs-B/16. Given that our model converges in just 50 epochs versus the typical 300 epochs in ViTs-based LNL methods, the total training time is reduced by approximately 6×, without sacrificing accuracy. In Table 3, we report the results for both RectMamba with ViTs and



(a) Symmetric 20% (EDKR)

(b) Symmetric 70% (EDKR)

(c) Asymmetric 20% (EDKR)

(d) Asymmetric 70% (EDKR)

(e) Symmetric 70% (LP)

(f) Asymmetric 70% (LP)

Fig. 7. T-SNE visualizations of test features from the CIFAR-10 dataset under different noise settings after 50 epochs of training. Subfigures (a)–(d) show results using RectMamba with EDKR under symmetric and asymmetric noise at 20 % and 70 %. Subfigures (e) and (f) show results of a controlled variant, where the EDKR module is replaced with Label Propagation (LP) under 70 % symmetric and asymmetric noise, respectively. Each of the ten classes—*Airplane*, *Automobile*, *Bird*, *Cat*, *Deer*, *Dog*, *Frog*, *Horse*, *Ship*, *Truck*—is distinctly represented by a unique color: deep blue, vibrant orange, grassy green, bold red, muted purple, earthy brown, soft pink, medium grey, bright lime green, and clear aqua, respectively.

CVMamba under symmetric and asymmetric noise on CIFAR-10 and CIFAR-100 datasets. The findings suggest that while both backbones—ViTs and CVMamba—demonstrate resilience to label noise, CVMamba consistently outperforms ViTs, particularly as the noise level increases. For example, on CIFAR-10 with 80 % symmetric noise, RectMamba with CVMamba achieves a test accuracy of 96.1 %, whereas the ViTs counterpart reaches 95.9 %. Similarly, under 40 % asymmetric noise on CIFAR-100, CVMamba achieves an accuracy of 82.1 %, surpassing ViTs's 79.2 %. This performance difference, although relatively small, emphasizes that the CVMamba backbone outperforms ViTs, which aligns with the visual state-space modeling approach of CVMamba. Specifically, the combination of dynamic selectivity and global receptive fields within the VSS blocks likely contributes to its superior handling of label noise across various noise scenarios.

5.3. T-SNE visulization

Fig. 7 presents T-SNE visualizations of network features derived from CIFAR-10 dataset test images after 50 epochs of training under various label noise conditions: symmetric at 20 % and 70 %, and asymmetric at 20 % and 70 %. Each plot illustrates clusters of points, color-coded to represent one of the ten CIFAR-10 classes: Airplane, Automobile, Bird, Cat, Deer, Dog, Frog, Horse, Ship, Truck.

In Fig. 7a-Symmetric 20 %, distinct class clusters are evident with minimal overlap, even at 20 % noise, underscoring RectMamba's capability to maintain accurate class distinctions. Despite the higher noise level, Fig. 7b-Symmetric 70 % displays well-defined clusters with slight blending, demonstrating the model's robust noise mitigation strategies.

Fig. 7c-Asymmetric 20 % retains clear class separation with minor overlaps, even under moderate asymmetric noise, highlighting strong feature extraction abilities. Lastly, Fig. 7d-Asymmetric 70 % shows more mixed clusters than at 20 % noise but still achieves significant class separation, showcasing the model's effectiveness in handling severe noise while preserving class integrity.

To further investigate local consistency, we replaced EDKR with Label Propagation (LP) [81] in RectMamba and visualized feature clusters under 70 % noise. As shown in Fig. 7e and Fig. 7f, LP results show noticeably more overlap between classes than EDKR, particularly under asymmetric noise. This indicates that LP, while enforcing smoothness, lacks the uncertainty filtering needed to prevent label diffusion across class boundaries, thus weakening robustness to label noise.

5.4. ANIMAL-10N

In our comprehensive study as referenced in Table 4, the effectiveness and prowess of our developed RectMamba method are quite evident. It strikingly exhibits its superiority by achieving an exceptional test accuracy of 92.5 % on the highly diverse ANIMAL-10N dataset. This performance spectacularly surpasses well-recognized and established techniques such as RegCE, SSR+, Jigsaw-ViT, and PASS. These aforementioned methods deliver test accuracies of 88.3 %, 88.5 %, 89.0 %, and 89.2 % respectively, which while commendable, are decisively outperformed by our RectMamba method. Notably, our RectMamba method also competitively edges out other strong performers like SANM and DivideMix + MC + SDA. Both of these methods deliver a test accuracy of over 89.0 %, which while notable, still falls short of the performance delivered by RectMamba.

5.5. Clothing1M

The Clothing1M dataset poses a unique challenge due to its complex mix of data noise that is neither purely symmetric nor entirely asymmetric, closely mimicking real-world data errors. The evaluation of the RectMamba method on this dataset highlights its robustness and effectiveness in dealing with label noise in practical scenarios. As illustrated in Table 5, RectMamba achieves an accuracy of 74.93 %, placing it among the top performers for this challenging dataset. While it falls just short of the highest accuracy—achieved by DivideMix* + SNSCL at 75.31 %—RectMamba's performance is remarkably close, with only a 0.38 % difference from the leading score. This minimal gap underscores RectMamba's proficiency in managing diverse and unpredictable noise patterns, a critical factor for real-world applications. This demonstrates that RectMamba's ability to handle noisy labels is highly competitive and practical for real-world data challenges, making it a valuable tool for addressing diverse and challenging noise patterns.

Table 4
Testing accuracy (%) on ANIMAL-10N.

Method	Test Accuracy
CE	79.4
SELFIE [80]	81.8
TCC-net [54]	83.2
PLC [99]	83.4
NCT [100]	84.1
InstanceGM [95]	84.6
OT-Filter [90]	85.5
SLRLNL [101]	86.4
CCE + HMW [55]	86.5
Dynamic Loss [102]	86.6
DISC [103]	87.1
RegCE [97]	88.3
SSR+ [96]	88.5
Robust LR [56]	88.5
Jigsaw-Vit [98]	89.0
SURE [104]	89.0
CWCL [105]	89.0
PASS [106]	89.2
SANM [107]	89.3
DivideMix + MC + SDA [108]	90.5
RectMamba (ours)	**92.5**

Table 5
Testing accuracy (%) on Clothing1M.

Method	Test Accuracy
CE	69.21
TCC-net [54]	70.46
TAkS [61]	70.28
Jigsaw-Vit [98]	72.49
SLRLNL [101]	74.15
C2D* [94]	74.84
ScanMix [109]	74.35
DivideMix* [32]	74.76
LongReMix [70]	74.38
InstanceGM [95]	74.40
OT-Filter [90]	74.50
CRAS [62]	74.54
DIST + DivideMix [103]	74.79
TCL + * [88]	74.85
ELR + * [86]	74.81
SSR+ [96]	74.83
UNICON* [89]	74.98
AugDesc* [93]	75.11
RML-Semi [57]	75.14
DivideMix* + SNSCL [91]	**75.31**
RectMamba (ours)	74.93

6. Challenges and future directions

While the proposed RectMamba method has demonstrated significant performance gains across various datasets, there remain certain limitations that need to be addressed for further advancements. Below, we outline the main challenges and propose future directions for enhancing the method's utility and robustness.

6.1. Challenges

Hyperparameter Sensitivity: A notable drawback of the KNN-based label correction mechanism is its sensitivity to hyperparameters, particularly the number of neighbors k. As shown in our experiments,

variations in k can impact performance, requiring careful tuning to achieve optimal results. This sensitivity makes the method less accessible to users who may lack experience in hyperparameter optimization.

High Label Noise in Similar Classes: In real-world datasets like Clothing1M, where visually similar classes (e.g., "Knitwear" and "Sweater") are prevalent, the model struggles to differentiate effectively. This results in reduced label correction accuracy, particularly when inter-class confusion is high. Despite improvements in noise robustness, label rectification in such cases remains challenging.

Lack of Granular Uncertainty Modeling: Although entropy-divergence captures prediction confidence, the current framework lacks explicit uncertainty modeling at the sample or region level. This can limit the model's ability to effectively filter hard-to-correct noisy samples effectively.

Handling Incomplete and Sparse Data: While RectMamba demonstrates strong robustness against label noise, it currently assumes fully observed and complete input data. In real-world applications, missing features or sparse observations frequently arise due to incomplete annotations, sensor dropout, or occlusions. This limitation may impair the performance of both the CVMamba feature extractor and the EDKR correction module, which rely on high-quality representations for clean sample selection.

Recent work, such as the Outlier-Resilient Autoencoder [110], introduces mechanisms for reconstructing missing high-dimensional data prior to classification. Similarly, the Three-Way Decision-Incorporated Online Sparse Feature Selection model [111] and Non-negative Latent Factor Analysis [112] provide effective solutions for sparsity-aware learning. These insights highlight a gap in RectMamba's current formulation.

6.2. Future directions

Automated Hyperparameter Optimization: Addressing hyperparameter sensitivity is another crucial direction for future work. By incorporating automated hyperparameter tuning techniques, such as Bayesian optimization or reinforcement learning, the dependency on manual tuning can be minimized. This would enhance the usability of RectMamba, particularly for users less familiar with hyperparameter selection, while also improving the efficiency of the label correction process.

Domain-Specific Pretraining and Self-Supervised Learning: To further improve performance in high-noise environments with similar classes, future work could explore domain-specific pretraining. By training the RectMamba model on datasets with characteristics similar to those of the target domain (e.g., clothing-related datasets for Clothing1M), the model may better adapt to specific features and noise patterns. Additionally, integrating self-supervised learning techniques to better leverage unlabeled data could enhance feature representation and robustness to label noise.

GBC-based Label Rectification: Granular-Ball Computing (GBC) [113,114] offers a promising direction. It models local uncertainty via granular-ball structures. Incorporating GBC-based classifiers like 3WC-GBNRS++ [115] could improve the precision of noisy label rectification. 3WC-GBNRS++ introduces a three-way decision mechanism based on minimum fuzziness loss, enabling finer handling of uncertain samples. By forming granular balls via a DBSCAN-based method and applying majority voting, this coarse-to-fine strategy provides a scalable and interpretable way to model uncertainty. Integrating such mechanisms into RectMamba could mitigate label confusion in high-noise, visually similar class settings, such as "Knitwear" vs. "Sweater" in Clothing1M, by suppressing unreliable corrections and focusing learning on confidently classified regions.

Future Integration of Missing Data Strategies: In future extensions of RectMamba, we plan to incorporate an autoencoder-based pre-processing stage for recovering incomplete visual or tabular inputs. This will enhance downstream feature consistency and reduce the negative impact of missing regions on similarity-based relabeling. Additionally, we will explore sparse-aware variants of the EDKR strategy, integrating confidence-driven neighbor filtering and low-rank latent factor modeling to mitigate the adverse effects of sparse sample distributions. These improvements are expected to bolster model robustness across incomplete, noisy, and high-dimensional datasets.

7. Conclusion

Our method, RectMamba, manages noisy labels, employing CVMamba, EDKR and Adaptivemix to maintain the purity of the training dataset. CVMamba resolves the warm-up phase challenges by counteracting noise memorization and the dependency on external data through convolutional vision Mamba pre-training, providing our model with the capabilities to understand intrinsic data patterns effectively and to possess high discriminative power. Entropy-Divergence, a cornerstone of RectMamba, eliminates the need for hyperparameter tuning by utilizing entropy and divergence to refine the datasets for superior classification performance. We also incorporated the KNN technique to improve the selection process for clean samples, securing a more trustworthy dataset. Our meticulous integration of Adaptivemix strategies and epoch-wise modulation of the correction ratio allows for a finely-tuned label correction process. Our methodology has proven to be highly effective, outperforming existing noise correction techniques on multiple benchmark datasets.

CRediT authorship contribution statement

Ningwei Wang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Weiqiang Jin:** Writing – review & editing, Visualization, Formal analysis. **Haixia Bi:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Formal analysis, Conceptualization. **Guang Yang:** Writing – review & editing, Funding acquisition.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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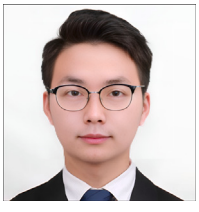
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Author biography



Ningwei Wang is currently a Ph.D. candidate in Electronic and Information Engineering at Xi'an Jiaotong University, Xi'an, China. He received a Software Engineering Master's degree from Beijing Union University. His research interests include remote sensing image processing and machine learning algorithms. His personal representative works are displayed on Github: https://github.com/ningwei-wang/.



Weiqiang Jin is currently a Ph.D. candidate in Electronic and Information Engineering at Xi'an Jiaotong University, Xi'an, China. He received a Computer Science Master's degree from Shanghai University. His main research interests are in natural language processing, representation learning, and reinforcement learning, especially the multi-agents information interaction and controlling. His personal representative works are displayed on Github: https://github.com/albert-jin/. And his personal website is located at https://albert-jin.github.io/.



Haixia Bi received the B.S. and M.S. degrees in computer science and technology from the Ocean University of China, Qingdao, China, in 2003 and 2006, respectively, and the Ph.D. degree in computer science and technology from Xi'an Jiaotong University, Xi'an, China, in 2018.,She was a Post-Doctoral Research Fellow with the University of Derby, Derby, U.K., from 2018 to 2019, and the University of Bristol, Bristol, U.K., from 2019 to 2021. She is currently an Associate Professor with the School of Information and Communication Engineering, Xi'an Jiaotong University. Her research interest include machine learning, remote sensing image processing, and digital health data processing.



Guang Yang is an Associate Professor (Senior Lecturer) in the Bioengineering Department and Imperial-X at Imperial College London. He holds a UKRI Future Leaders Fellowship and serves as an Honorary Senior Lecturer in the School of Biomedical Engineering & Imaging Sciences at King's College London. He is an Associate Editor of IEEE Transactions and npj Digital Medicine. His research group is dedicated to developing novel and translational techniques for imaging and biomedical data analysis. The group's focus encompasses research and development in data-driven fast imaging, data harmonization, data synthesis, federated learning, explainable AI, and AI in drug discovery. Currently, his work spans a wide range of clinical applications in ageing, cardiovascular disease, lung disease, and oncology. For more information about Yang's Lab, visit: https://www.yanglab.fyi/ and Twitter: @gyangMedIA.